LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 8: Cryptography: The Caesar Cipher

Module 3 | 45 minutes | Ciphers and modular arithmetic | Grades 6–8 · ages 12–14

Objective: Encrypt and decrypt messages with a Caesar cipher using letter positions and modular arithmetic.

Success criteria

- I can encrypt and decrypt with a Caesar cipher.
- I can use mod to wrap around.
- I can explain why it's easy to break.

Key vocabulary

- **cipher**: A method for encrypting messages.
- **key**: The secret value used to encrypt and decrypt.
- **mod**: The remainder after division; wraps numbers around.
- **brute force**: Trying every possible key.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Cipher wheels (5 min): paper wheels; pairs exchange shifted messages.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Encrypt a word: learners follow the steps.
14 min	**Perform**	Code Lab: encrypt, Wrap around
9 min	**Probe**	Code Lab: decrypt, activity
4 min	**Wrap-up**	Exit ticket: With shift 5, what is "a" encrypted to?

Unplugged warm-up

Cipher wheels (5 min): paper wheels; pairs exchange shifted messages.

Teaching the idea

Caesar cipher: position lookup (linear search over the alphabet), modular addition for wrap-around, decryption by inverse shift. Discuss brute force and frequency analysis; contrast with modern key spaces.

Analogy: A clock: after 12 comes 1 — that's mod.

Live demo: Encrypt "xyz" on a cipher wheel and in code to show wrap-around.

Questions to ask

Ask	Listen for
What is the key in a Caesar cipher?	The shift.
Why do we need mod?	To wrap from z back to a.
How many keys are there?	25 useful shifts.

Differentiation

Support
Provide the position-finding loop.

Stretch
Brute-force: print all 25 shifts of a secret message.

Common misconceptions

Misconception	How to fix it
Mod only works for positive numbers	Scratch-style mod wraps negatives too.
A secret method is enough	Security relies on the key, not hiding the method.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
With shift 5, what is "a" encrypted to?

Homework
Send a family member a Caesar-encrypted message and tell them the key.

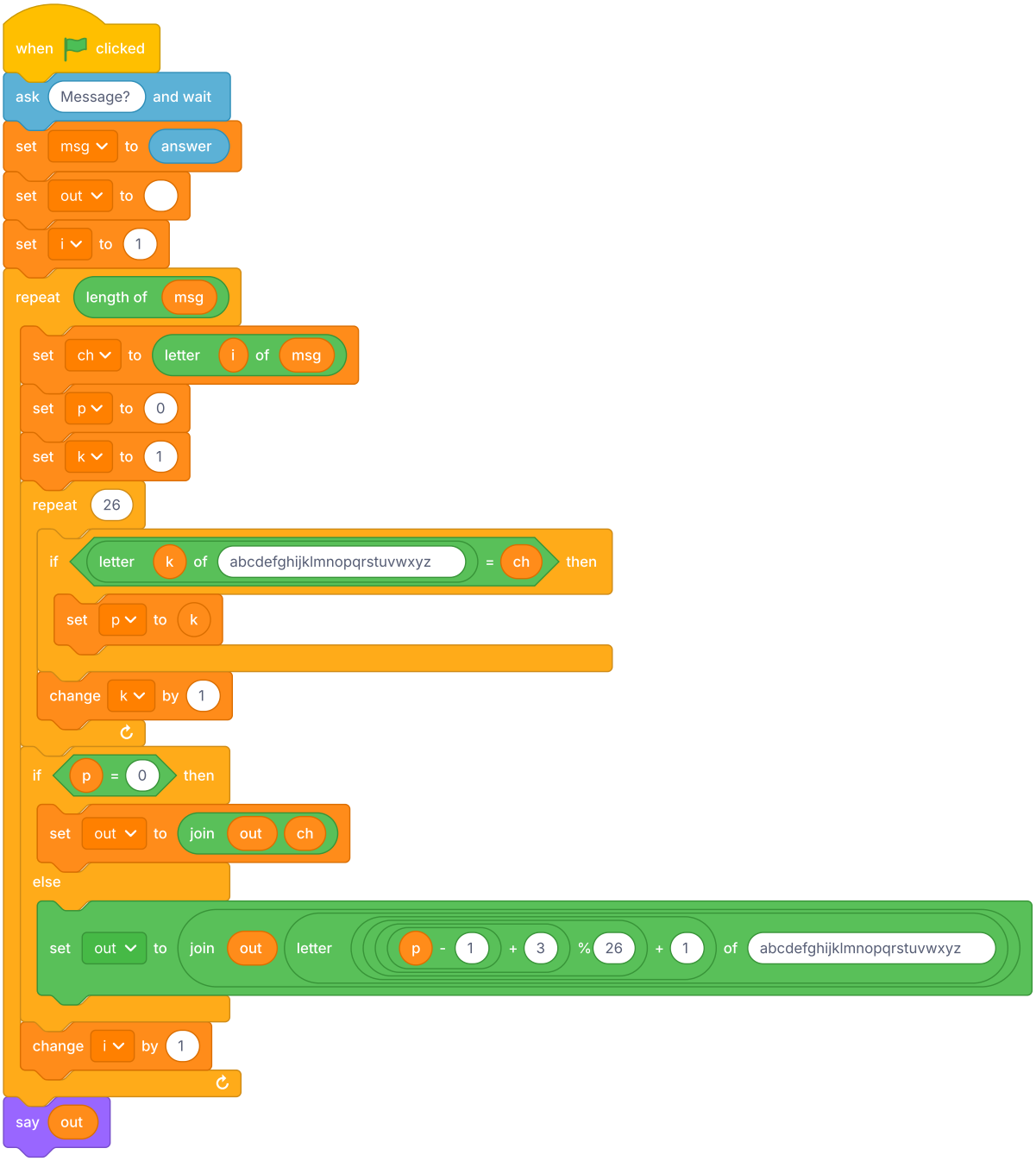
Answer key — Kitabash activities

1. Code Lab: encrypt (perform · code lab)

Goals checked: hello → khoor · xyz wraps to abc · Uses mod

One solution:

Agent



2. Wrap around (perform · multiple choice)

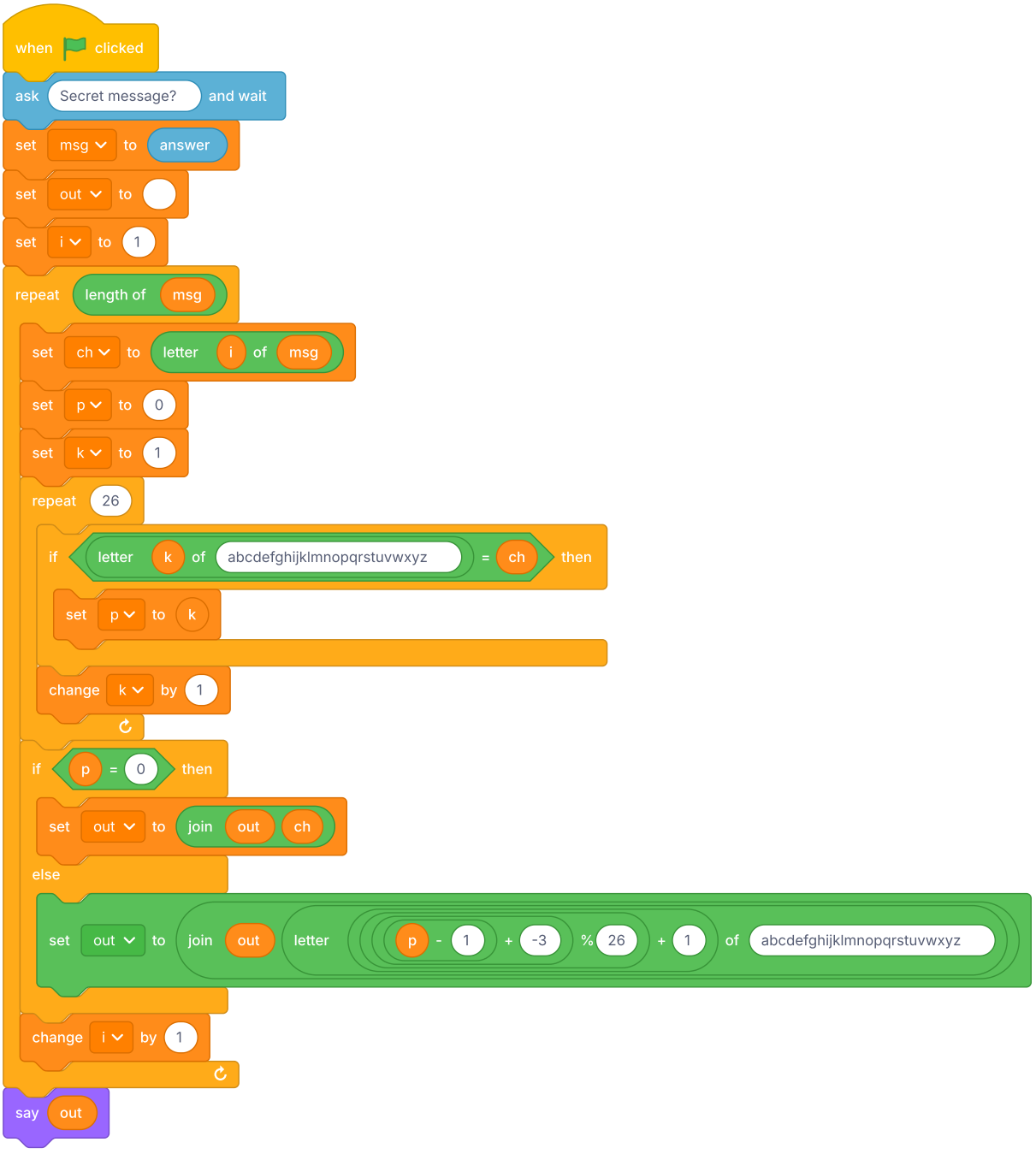
Answer: b

3. Code Lab: decrypt (probe · code lab)

Goals checked: khoor → hello · abc wraps to xyz

One solution:

Agent



4. Explain your thinking (probe · free response)

Look for: shift, try, key. **Example:** There are only 25 possible shifts, so an attacker can just try them all until the message makes sense. They could also look for the most common letter, which is probably e. Modern encryption uses keys with so many possibilities that trying them all would take millions of years.

Answer key — worksheet

1. **Key words:** cipher = A method for encrypting messages. · key = The secret value used to encrypt and decrypt. · mod = The remainder after division; wraps numbers around. · brute force = Trying every possible key.
2. **Plan your code:** Goals: hello → khoor; xyz wraps to abc; Uses mod.
3. **Wrap around:** A. b — y → z → a → b (wrapping around after z).
4. **Plan your code:** Goals: khoor → hello; abc wraps to xyz.
5. **Explain your thinking:** There are only 25 possible shifts, so an attacker can just try them all until the message makes sense. They could also look for the most common letter, which is probably e. Modern encryption uses keys with so many possibilities that trying them all would take millions of years.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.